Sums of Even Numbers

A synthetic example

Theorem 1. *The sum of two even natural numbers is even.*

Theorem 2. *The sum of four even natural numbers is even.*

Theorem 3. *The sum of three even natural numbers is even.*

Theorem 4. *Twice any number is even.*

Notation 1. *We write E for the set of even natural numbers.*

Theorem 5. *The sum of two even natural numbers is even.*

Proof. Write the numbers as $2a$ and $2b$. Then

$$2a + 2b = 2(a + b). \tag{1}$$

□